INCH-POUND
MIL-DTL-8416C
13 May 2005
SUPERSEDING
MIL-A-8416B (USAF)
18 October 1979

DETAIL SPECIFICATION

ADAPTER, HEADSET-MICROPHONE MX-1646( )/AIC

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of adapter designated Adapter, Headset-Microphone MX-1646( )/AIC. This adapter permits the use of low impedance dynamic headsets and microphones with equipment designed for use with 600-ohm headsets and carbon microphones.

1.1.1 Types. In the type designation, the parentheses will be either removed or replaced by a letter. This information will be furnished after each contract has been awarded upon application by the contractor to the procuring activity. The complete nomenclature or type number will be used on nameplates, shipping records, and in instruction books as applicable.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-9177	Connector, Audio, Airborne, General Specification For
MIL-DTL-9177/3	Connector, Audio, Airborne, Jack, Cable, 4 Contact
MIL-DTL-5898	Cords, Electrical (WF-5/U, WF-14/U, WD-34/U, WM-53/U, WM-85/U)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1285	Marking of Electrical and Electronic Parts

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(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL ORGANIZATION for STANDARDIZATION (ISO)

ISO 10012-1	-	Quality Assurance Requirements For Measuring Equipment Part 1: Metrological Confirmation System for Measuring Equipment.

(Copies of these documents are available from http://www.iso.ch or from the International Organization for Standardization American National Standards Institute, 11 West 42nd Street, 13th Floor, New York, NY 10036.)

NATIONAL CONFERENCE OF STANDARDS LABORATORIES (NCLS)

NCSL Z540-1	-	General Requirements for Calibration Laboratories and Measuring and Test Equipment.

Copies of these documents are available from http://www.ncsli.org or from the National Conference of Standards Laboratories (NCSL), 2995 Wilderness Place, Suite 107, Boulder, CO 80301-5404.)

IPC - ASSOCIATION CONNECTING ELECTRONICS INDUSTRIES

J-STD-006	-	Requirements for Electronic Grade Solder Alloys and Fluxed and Non-Fluxed Solid Solders for Electronic Soldering Applications.

(Copies of these documents are available from http://www.ipc.org or from the IPC - Association Connecting Electronics Industries, 3000 Lakeside Drive, Suite 309 S, Bannockburn, IL 60015-1249.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AS21919	Clamp, Loop Type, Cushioned Support

(Copies of these documents are available online from http://www.sae.org or from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.4.

3.2 Materials. Materials shall be as specified herein. When a definite material is not specified, a material shall be used that will enable the adapters to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.2.1 Metals. Metals shall be corrosion resistant and of the proper alloy and hardness necessary to provide the required strength and rigidity to meet the performance requirements of this specification.

3.2.2 Solder. Solder shall conform to composition Sn60Pb40 in accordance with J-STD-006.

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3.2.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design.

3.3.1 Connector. The connector shall be in accordance with MIL-C-9177 and MIL-DTL-9177/3 for the replacement for the U-92 A/U Audio Connector.

3.3.2 Electrical cords. Electrical cords WF-14/U, as shown on figure 1, shall be in accordance with MIL-DTL-5898.

3.3.3 Adapter. The adapter shall electrically match an 8-Ohm headset to a 600-ohm source and use an amplifier to couple a 5-Ohm dynamic microphone into input circuits designed for military carbon microphones. Configuration is shown on figure 1. Power for the amplifier shall be obtained from the carbon microphone input circuit in the aircraft when the unterminated cord of the adapter is connected to the existing carbon microphone terminals. Hermetic sealing or encapsulation may be employed to meet the environmental requirements of this specification.

3.3.4 Schematic diagram. The adapter shall be wired as shown on figure 2.

3.3.5 Automatic gain control (AGC). The adapter shall have incorporated AGC within the adapter container.

3.3.6 Mounting. The adapter shall be designed to be mounted by two type AS21919WD(*)22 clamps spaced on two inch centers. The two clamps in accordance with AS21919 shall be supplied with or as a part of each adapter.

3.3.7 Supply voltage. The adapter shall be capable of normal operation over a supply voltage range of 5-30 volts direct current (VDC).

3.3.8 Shielding. Cord shields shall be connected to the housing from the headset-matching device and microphone adapter. The adapter exterior housing shall be electrically grounded.

3.3.9 Weight. Weight shall not exceed 6 ounces excluding cords and connectors.

3.3.10 Pure tin. The use of pure tin is prohibited both internally and externally. Tin content of adapter components and solder shall not exceed 97 percent. Tin shall be alloyed with a minimum of 3 percent lead (see 6.5).

3.4 Performance characteristics.

3.4.1 Noise. Output background noise, with no signal input, shall be at least 50 dB below normal output when the adapter is placed in any position within 10 inches of an isolated, insulated conductor carrying 10 amperes of 400 Hz current. Noise output due to ripple within the range of 100-8000 Hz shall not exceed the value obtained with switch SW-1 on figure 3 “OFF” position versus when measured in the “ON” position (see 4.6.2).

3.4.2 Distortion. Harmonic distortion shall not exceed 5 percent when operating at normal output levels. There shall be no oscillations or regeneration when Switch SW-2 on figure 3 is placed in position 1 (see 4.6.3).

3.4.3 Current consumption. The total current drain of the adapter shall not exceed 25 milliamperes, direct current (DC) when SW-1 on figure 3 is in the “ON” position (see 4.6.4).

3.4.4 Frequency response. The adapter shall provide a gain of 62 ±3 dB over the frequency range of 300 to 6000 Hz with various load impedance of 50, 100, 200, 300, 400 and 500-ohms (see 4.6.5).

3.5 Environmental.

3.5.1 Cable retention. The cords shall withstand a 25 pound pull in line with the adapter body and not pull free from the housing of cause an electrical discontinuity (see 4.6.6).

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3.5.2 Temperature range. The adapter shall remain operational throughout the range of -55ºC to +85ºC and show no damage as a result of exposure to the temperature range specified herein (see 4.6.7).

3.5.3 Humidity. The insulation resistance between the terminals of the cords and the case shall be at least 100 megohms (see 4.6.8).

3.5.4 Altitude. The adapter case shall not deform in a way to cause electrical performance degradation (see 4.6.9).

3.5.5 Salt atmosphere. There shall be no excessive corrosion of metal parts (see 4.6.10).

3.5.6 Vibration. There shall be no loosening of parts or separation of parts (see 4.6.11).

3.5.7 Shock (specified pulse). There shall be no loosening or separation of parts (see 4.6.12)

3.5.8 Fungus. The product shall be constructed of fungus-inert materials, and shall not exhibit any degradation in performance due to prolonged exposure to fungus-producing conditions. Following such exposure, the product shall exhibit the following performance (see 4.6.13):

Visual examination:	No evidence of physical damage which would cause a malfunction, and no visible fungus or corrosion.

3.6 Finish. Adapters shall be painted lusterless black.

3.7 Marking. Adapters shall be marked in accordance with MIL-STD-1285 with the type designation and manufacturer's code symbol or name. In addition, a green paint dot shall be applied to the case of the adapter near the point of exit of the cord, which is not terminated with Audio Connector M9177/3-1. The diameter of the dot shall be between .125 and .250 inch. The un-terminated cord shall have a tag attached bearing the following note:

"CAUTION: Red lead, negative; Green lead, positive"

3.8 Workmanship. Adapters shall be processed in such a manner as to be uniform in quality and shall be free from loose or deposited foreign materials, and other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Conformance inspection (see 4.5).

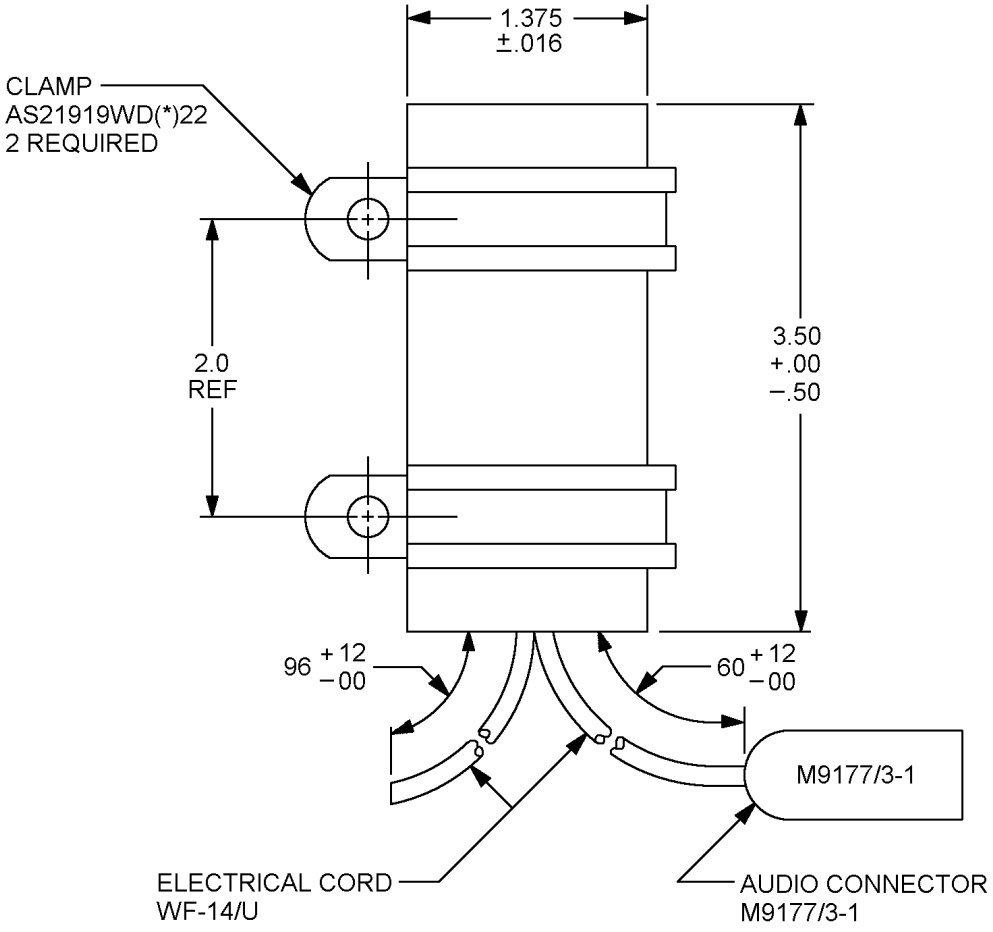
(1) Inspection of product for delivery (see 4.5.1).

4.2 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of sufficient accuracy, quality and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government. Calibration of the standards which control the accuracy of inspection equipment shall be in accordance with ISO 10012-1 and NCSL Z540-1.

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-202, section 2.

4.4 First article inspection. When required by the procurement activity, first article inspection shall be performed by the suppliers, after award of contract and prior to production, at a laboratory acceptable to the government.

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Inches	mm	Inches	mm
.016	0.41	3.50	88.9
.50	12.7	12	304.8
1.375	34.93	60	1524
2.0	50.8	96	2438.4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.

FIGURE 1. Adapter, headset-microphone MX-1646( )/AIC.

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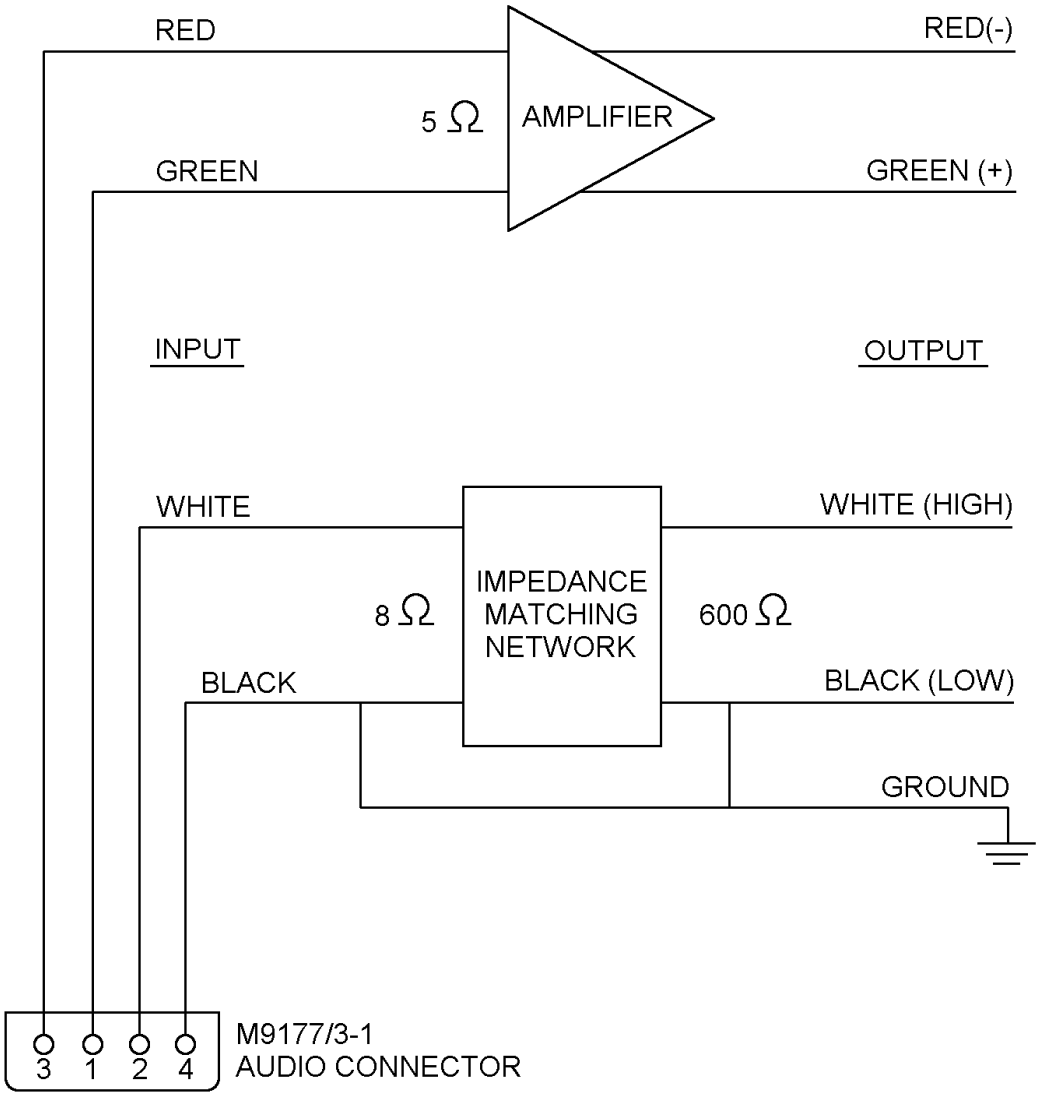


FIGURE 2. Schematic, headset-microphone adapter MX-1646( )/AIC.

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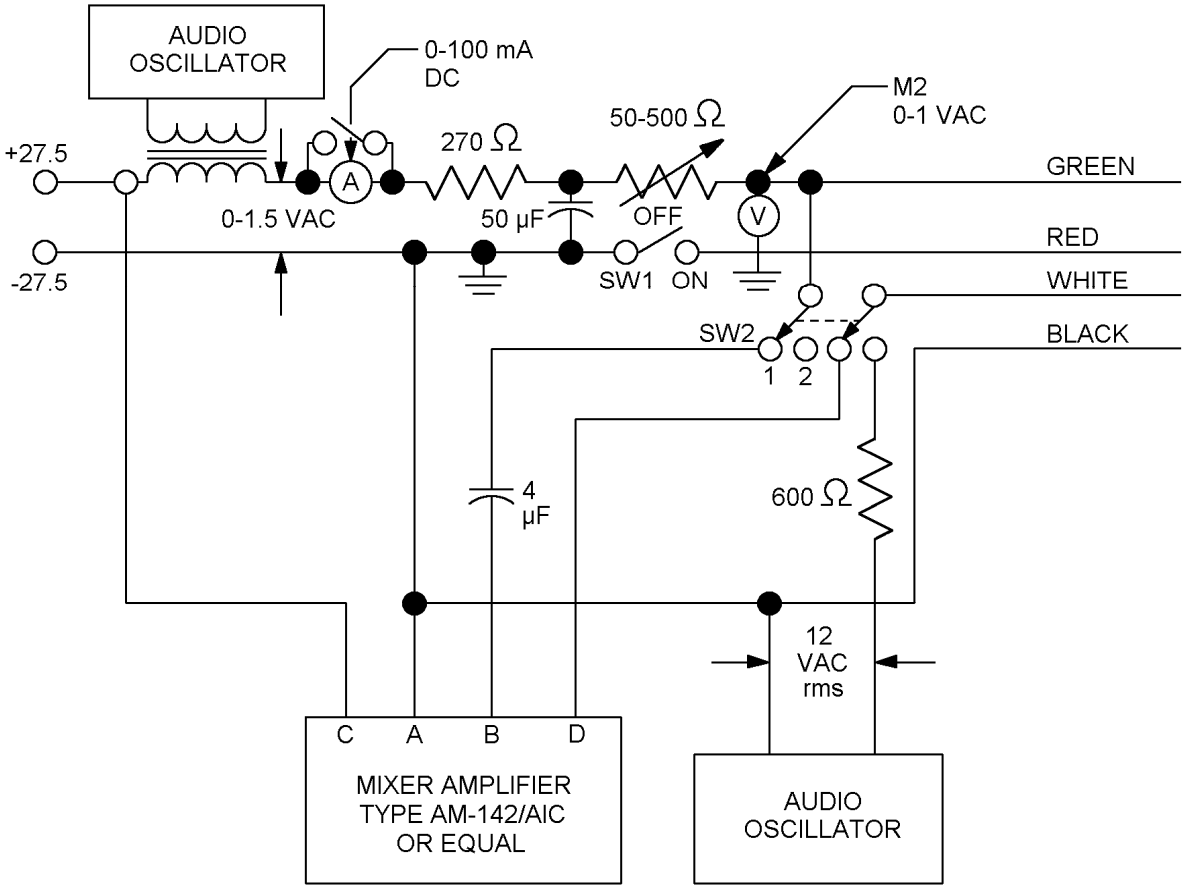


FIGURE 3. Interphone simulator.

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4.4.1 Sample size. Six (6) adapters shall be submitted for first article inspection as required in table I.

4.4.2 Test routine. Sample units shall be subjected to the inspections specified in table I in the order shown below.

TABLE I. First article inspection.

Examination or test	Requirement paragraph	Method paragraph	No. of units
Visual and mechanical	3.2, 3.3, 3.6, 3.7, 3.8	4.6.1	All
Noise	3.4.1	4.6.2	
Distortion	3.4.2	4.6.3	
Current consumption	3.4.3	4.6.4	
Frequency response	3.4.4	4.6.5	
Cable retention	3.5.1	4.6.6	
Temperature range	3.5.2	4.6.7	
Humidity	3.5.3	4.6.8	
Altitude	3.5.4	4.6.9	
Salt spray	3.5.5	4.6.10	
Vibration	3.5.6	4.6.11	
Shock (specified pulse)	3.5.7	4.6.12	

4.5 Conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspections.

4.5.1.1 Inspection lot. An inspection lot shall consist of all adapters covered by this specification, produced under essentially the same conditions and offered for inspection at one time.

4.5.1.2 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot, rework it to correct the defect, or screen out the defective units, as applicable and re-inspect. Such lots shall be separate from new lots, and shall be clearly identified as re-inspected lots.

4.5.1.3 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table II.

TABLE II. Group A inspection.

Examination or test	Requirement paragraph	Method Paragraph
Visual and mechanical	3.2, 3.3, 3.6, 3.7, 3.8	4.6.1
Cable retention	3.5.1	4.6.6
Noise	3.4.1	4.6.2
Distortion	3.4.2	4.6.3
Frequency response	3.4.4	4.6.5

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4.5.1.4 Sampling plan. A sample of parts shall be randomly selected in accordance with table III, if one or more defects are found, the lot shall be re-screened and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected in accordance with table III, if one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

TABLE III. Sampling plan.

Lot size	Sample size
2 to 8	1/
9 to 15	13 1/
16 to 25	13
26 to 50	13
51 to 90	13
91 to 150	13
151 to 280	20
281 to 500	29
501 to 1200	34
1201 to 3200	42

1/ Indicates entire lot must be inspected or sample size not to exceed lot size.

4.5.2 Inspection of packaging. Sample packages or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with section 5.

4.5.3 Sampling plan. Accept on zero, for general inspection as specified in table II.

4.6 Methods of examination and test.

4.6.1 Visual and mechanical examination. The adapters shall be examined to verify that the materials, design, construction, physical dimensions, finish, marking, and workmanship are as specified herein (see table I).

4.6.2 Noise. Measure and record the output background noise, with no signal input to the adapter when the adapter is placed within 10 inches of the required conductor (see 3.4.1).

4.6.3 Distortion. The harmonic distortion shall be measured and recorded and shall not exceed the value specified herein (see 3.4.2).

4.6.4 Current consumption. The current drain shall be measured at the output of the primary power supply on figure 3 and shall not exceed the value specified (see 3.4.3).

4.6.5 Frequency response. The frequency response of the adapter shall be measured at 300, 1000, 2000 and 6000 Hz and shall fall within the limits specified (see 3.4.4).

4.6.6 Cable retention. A 25 pound weight shall be hung vertically along the axis of each cord and the adapter for 5 minutes. A visual inspection shall be made for breakage or damage failures (see 3.5.1).

4.6.7 Temperature range. The adapter shall be placed in a test chamber at 85ºC for four (4) hours. It shall be tested for normal operation and inspected for damage within 15 minutes of the end of the four (4) hours. It shall then be placed in a test chamber at -55ºC for ten (10) hours. Upon completion of the cold soak, the adapter shall be examined for physical damage (including the cords) and tested for normal operation (see 3.5.2).

4.6.8 Humidity. Humidity testing shall be in accordance with MIL-STD-202, method 103, test condition B. At the conclusion of this test, the insulation resistance between the metal case and all of the conductors shall be at least 100 megohms (see 3.5.3).

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4.6.9 Altitude. Altitude testing shall be in accordance with MIL-STD-202, method 105, test condition A, unmounted and subjected to visual inspection upon completion (see 3.5.4).

4.6.10 Salt atmosphere. Salt spray test shall be in accordance with MIL-STD-202, method 101, test condition B. No excessive corrosion shall be present following the test (see 3.5.5).

4.6.11 Vibration. Vibration of the adapter excluding cords and connectors shall be in accordance with MIL-STD-202, method 201 and checked for normal operation (see 3.5.6).

4.6.12 Shock. Shock test shall be in accordance with MIL-STD-202, method 213, test condition C and visually inspected (see 3.5.7).

4.6.13 Fungus. The contractor shall certify that the materials used meet the requirements for fungus-inert materials in table IV, or test in accordance with method 508 of MIL-STD-810 for a period of 28 days. The following details shall apply (see 3.5.8):

a. The complete assembly shall be tested.
b. Within 1 hour after the test the components shall be examined for evidence of physical damage.

TABLE IV. Fungus-inert materials.

Group I – Fungus inert materials (Fungus-inert in all modified states and grades)	
Acrylics	1/ Polyamide
Acrylonitrile-styrene	Polycarbonate
Acrylonitrile-vinyl-chloride copolymer	Polyester-glass fiber laminates
Asbestos	Polyethylene, high density
Ceramics	(above 0.940)
	Polyethylene terephthalate
Chlorinated polyester	
Fluorinated ethylenepropylene copolymer	Polyamide
(FEP)	Polymonochlorotrifluoroethylene
Glass	Polypropylene
Metals	Polystyrene
Mica	Polysulfone
Plastic laminates:	Polytetrafluoroethylene
Silicone-glass fiber	Polyvinylidene chloride
Phenolic-nylon fiber	Silicone resin
Diallyl phthalate	Siloxane-polyolefin polymer
Polyacrylonitrile	Siloxane polystyrene

1/ Literature shows that under certain conditions polymides may be attacked by selective micro-organisms. However, for military applications, they are considered group I.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's System Commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Headset-Microphone Adapter MX-1646( )/AIC is for the purpose of adapting low impedance dynamic headsets and microphones to communication equipment designed for use with 600 ohm headsets and microphones.

6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification (see 3.1).
b. Levels of preservation – packaging and packing required (see 5.1).
c. Special marking, if required (see 5.1).
d. When first article is required, based upon contractor performance history (see 3.1).

6.3 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. Table V lists the Environmental Protection Agency (EPA) top seventeen hazardous materials targeted for major usage reduction. Use of these materials should be minimized or eliminated unless needed to meet the requirements specified herein (see section 3).

TABLE V. EPA top seventeen hazardous materials.

Benzene	Dichloromethane	Tetrachloroethylene
Cadmium and Compounds	Lead and Compounds	Toluene
Carbon Tetrachloride	Mercury and Compounds	1,1,1 - Trichoroethane
Chloroform	Methyl Ethyl Ketone	Trichloroethylene
Chromium and Compounds	Methyl Isobutyl Ketone	Xylenes
Cyanide and Compounds	Nickel and Compounds	

6.4 Subject term (key word) listing.

AGC
Audio
Clamp
Connector
Distortion
Dynamic
Noise
Schematic
Solder
Voltage
Whisker

6.5 Tin whisker growth. The use of alloys with tin content greater than 97 percent may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture, and can develop under typical operating conditions on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead have shown to inhibit the growth of tin whiskers (see 3.3.10).

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Air Force – 11
DLA – CC

Preparing activity:
DLA – CC

Review activity:
Air Force – 99

(Project 5965-0354-000)

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